Cambridge Assessment International Education

PAPERDOME

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

0815246258

FURTHER MATHEMATICS **9231/42**

Paper 4 Further Probability & Statistics **October/November 2023**

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

DC (EF/FC) 316301/1

[Turn over

BLANK PAGE

for both $n \geq 30$

1 A factory produces small bottles of natural spring water. Two different machines, X and Y, are used to fill empty bottles with the water. A quality control engineer checks the volumes of water in the bottles filled by each of the machines. He chooses a random sample of 60 bottles filled by machine X and a random sample of 75 bottles filled by machine Y. The volumes of water, x and y respectively, in millilitres, are summarised as follows.

$$\Sigma x = 6345 \quad \Sigma(x-\bar{x})^2 = 243.8 \quad \Sigma y = 7614 \quad \Sigma(y-\bar{y})^2 = 384.9$$

$\bar{x}$ and $\bar{y}$ are the sample means of the volume of water in the bottles filled by machines X and Y respectively.

Find a 95% confidence interval for the difference between the mean volume of water in bottles filled by machine X and the mean volume of water in bottles filled by machine Y. [6]

$(\bar{x}-\bar{y}) \pm z_{\frac{\alpha}{2}}\sqrt{\frac{s^2_x}{n_x} + \frac{s^2_y}{n_y}}$

$\Sigma(x-\bar{x})^2 \rightarrow \Sigma x^2 - \frac{(\Sigma x)^2}{n}$

$\bar{x} = \frac{6345}{60} = 105.75 \qquad \bar{y} = \frac{7614}{75} = 101.52$

$s^2_x = \frac{1}{59}(243.8) = 4.1322$

$s^2_y = \frac{1}{74}(384.9) = 5.201$

$z_{\frac{\alpha}{2}} = z_{0.975}$ where $100(\alpha - 1) = 95$

$C.I \rightarrow (105.75 - 101.52) \pm 1.96\sqrt{\frac{4.1322}{60} + \frac{5.201}{75}}$

$\rightarrow 4.23 \pm (1.96 \times 0.3718)$

$\rightarrow [3.50, 4.96]$

2 The number of breakdowns on a particular section of road is recorded each day over a period of 90 days. It is suggested that the number of breakdowns follows a Poisson distribution with mean 3.5. The data is summarised in the table, together with some of the expected frequencies resulting from the suggested Poisson distribution.

Number of breakdowns per day	0	1	2	3	4	5	6	7	8 or more
Observed frequency	0	5	13	17	21	16	9	5	4
Expected frequency	2.718	9.512	16.646	19.421	16.993	11.895	6.939	3.469	2.407

(a) Complete the table. [2]

$E(3) = e^{-3.5} \times \frac{3.5^3}{3!} \times 90 = 19.421$

$E(6) = e^{-3.5} \times \frac{3.5^6}{6!} \times 90 = 6.939$

(b) Carry out a goodness of fit test, at the 10% significance level, to determine whether or not Po(3.5) is a good fit to the data. [6]

H_0 : Po(3.5) is a good fit to the data

H_1 : Po(3.5) is not a good fit for the data

Combine first two and last two columns so that Expected values ≥ 5

	0-1	2	3	4	5	6	7 or more
obs	5	13	17	21	16	9	9
exp	12.23	16.646	19.421	16.993	11.895	6.939	5.876

test statistic $\rightarrow \Sigma \frac{(O-E)^2}{E}$

$\rightarrow \frac{(5-12.23)^2}{12.23} + \ldots\ldots$

$\rightarrow 4.274 + 0.798 + 0.3018 + 0.944$
$+ 1.417 + 0.612 + 1.661$
$= 10.0088$

$\rightarrow$ no estimated parameters

degrees of freedom $= n - v - 1$
$7 - 0 - 1$
$= 6$

$\chi^2_6(0.90)$
critical value $\rightarrow 10.64$

$10.0088 < 10.64 \rightarrow$ accept H_0

insufficient evidence to suggest that Po(3.5) is not a good fit for the data

3 Toby has a bag which contains 6 red marbles and 3 green marbles. He randomly chooses 3 marbles from the bag, without replacement. The random variable X is the number of red marbles that Toby obtains.

(a) Find the probability generating function of X. [3]

$P(R=0) = \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} = \frac{1}{84}$

$P(R=1) = \frac{6}{9} \times \frac{3}{8} \times \frac{2}{7} \times \frac{3!}{2!} = \frac{3}{14}$

$P(R=2) = \frac{6}{9} \times \frac{5}{8} \times \frac{3}{7} \times \frac{3!}{2!} = \frac{15}{28}$

$P(R=3) = \frac{6}{9} \times \frac{5}{8} \times \frac{4}{7} = \frac{5}{21}$

$G_X(t) = \frac{1}{84} + \frac{3}{14}t + \frac{15}{28}t^2 + \frac{5}{21}t^3$

Ling also has a bag which contains 6 red marbles and 3 green marbles. He randomly chooses 2 marbles from his bag, without replacement. The random variable Y is the number of red marbles that Ling obtains. It is given that the probability generating function of Y is $\frac{1}{12}(1+6t+5t^2)$.

The random variable Z is the total number of red marbles that Toby and Ling obtain.

(b) Find the probability generating function of Z, expressing your answer as a polynomial in t. [3]

$G_X(t) = \frac{1}{84}(1 + 18t + 45t^2 + 20t^3)$

$G_Z(t) = \frac{1}{84} \times \frac{1}{12}(1+6t+5t^2)(1+18t+45t^2+20t^3)$

$\rightarrow \frac{1}{1008}(1 + 18t + 45t^2 + 20t^3 + 6t + 108t^2 + 270t^3 + 120t^4 + 5t^2 + 90t^3 + 225t^4 + 100t^5$

$= \frac{1}{1008}(1 + 24t + 158t^2 + 380t^3 + 345t^4 + 100t^5)$

$$\frac{1}{1008}\left(1 + 158t^2 + 24t + 380t^3 + 345t^4 + 100t^5\right)$$

(c) Use the probability generating function of Z to find Var(Z). [4]

$$\text{Var}(X) = G''_X(1) + G'_X(1) - (G'_X(1))^2$$

$$G'_X(t) = \frac{1}{1008}\left(24 + 316t + 1140t^2 + 1380t^3 + 500t^4\right)$$

$$G'_X(1) = \frac{3360}{1008} = \frac{10}{3}$$

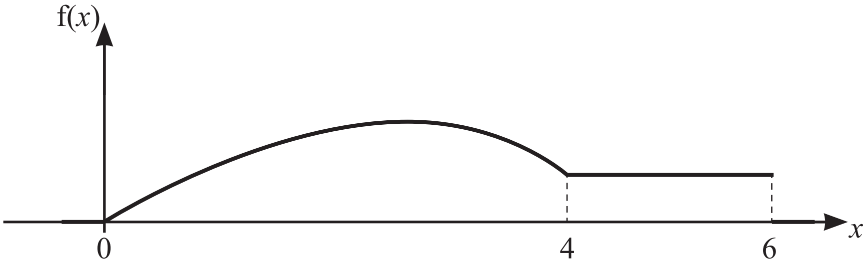
$$G''_X(t) = \frac{1}{1008}\left(316 + 2280t + 4140t^2 + 2000t^3\right)$$

$$G''_X(1) = \frac{26}{3}$$

$$\therefore \text{Var}(X) = \frac{26}{3} + \frac{10}{3} - \left(\frac{10}{3}\right)^2$$

$$= \boxed{\frac{8}{9}}$$

4



The diagram shows the continuous random variable X with probability density function f given by

$$f(x) = \begin{cases} \frac{1}{128}(4ax - bx^3) & 0 \leqslant x \leqslant 4, \\ c & 4 \leqslant x \leqslant 6, \\ 0 & \text{otherwise,} \end{cases}$$

where a, b and c are constants.

The upper quartile of X is equal to 4.

(a) Show that $c = \frac{1}{8}$ and find the values of a and b. [4]

$c \times (6-4) = 0.25$ → as total amount of area of PDF after UQ → 0.25

$C = 1/8$

at $x = 4$ $f(x) = \frac{1}{8}$

$\frac{1}{128}(16a - 64b) = \frac{1}{8}$ → $\boxed{a - 4b = 1}$

$\frac{1}{128}\int_0^4 4ax - bx^3 \, dx = 0.75$

$\left[2ax^2 - \frac{b}{4}x^4\right]_0^4 = 96$

$32a - 64b = 96$ → $a - 2b = 3$

$- \; a - 4b = 1$

$2b = 2$

$\boxed{a = 5}$ $\boxed{b = 1}$

(b) Find the exact value of the median of X. [3]

median in range $\to 0 \le x \le 4$

$\frac{1}{128}\int_0^m 20x - x^3 \, dx = 0.5$

$\frac{1}{128}\left[10x^2 - \frac{x^4}{4}\right]_0^m = 0.5$

$10m^2 - \frac{m^4}{4} = 64$

$m^4 - 40m^2 + 256 = 0$

$(m^2 - 32)(m^2 - 8) = 0$ $\quad m = \sqrt{8}$

$m = 2\sqrt{2}$

(c) Find $E(\sqrt{X})$, giving your answer correct to 2 decimal places. [3]

$E(\sqrt{x}) \to \int \sqrt{x}\, f(x) \, dx$

$\frac{1}{128}\int_0^4 20x^{3/2} - x^{7/2} \, dx + \frac{1}{8}\int_4^6 x^{1/2} \, dx$

$\frac{1}{128}\left[8x^{5/2} - \frac{2}{9}x^{9/2}\right]_0^4 + \frac{1}{8}\left[\frac{2}{3}x^{3/2}\right]_4^6$

$\to \frac{1}{128}\left[\frac{1280}{9}\right] + \frac{1}{8}\left[4.4646\right] = \boxed{1.67}$

5 A company is deciding which of two machines, X and Y, can make a certain type of electrical component more quickly. The times taken, in minutes, to make one component of this type are recorded for a random sample of 8 components made by machine X and a random sample of 9 components made by machine Y. These times are as follows.

Machine X	4.0	4.6	4.7	4.8	5.0	5.2	5.6	5.8	
Machine Y	4.5	4.9	5.1	5.3	5.4	5.7	5.9	6.3	6.4

The manager claims that on average the time taken by machine X to make one component is less than that taken by machine Y.

(a) Carry out a Wilcoxon rank-sum test at the 5% significance level to test whether the manager's claim is supported by the data. [6]

X	Rank	Y	Rank
4.0	1	4.5	2
4.6	3	4.9	6
4.7	4	5.1	8
4.8	5	5.3	10
5.0	7	5.4	11
5.2	9	5.7	13
5.6	12	5.9	15
5.8	14	6.3	16
		6.4	17
	55		98

M N

$m \leq n$

$H_0: m_x = m_y$

$H_1: m_x < m_y$

$W = \min(R_m, m(m+n+1) - R_m)$
$= \min(55, 89)$
$= 55$

critical value $\rightarrow m = 8, n = 9$, 5% one-tailed
$\rightarrow 54$

$55 > 54 \rightarrow$ accept H_0
$\rightarrow$ insufficient evidence to support manager's claim

(b) Assuming that the times taken to produce the components by the two machines are normally distributed with equal variances, carry out a t-test at the 5% significance level to test whether the manager's claim is supported by the data. [9]

$H_0: \mu_x = \mu_y \quad H_1: \mu_x < \mu_y$

$\Sigma x = 39.7 \quad \Sigma y = 49.5$

$\Sigma x^2 = 199.33 \quad \Sigma y^2 = 275.47$

$s_p^2 = \dfrac{S_{xx} + S_{yy}}{n_x + n_y - 2} \quad S_{xx} = 199.33 - \dfrac{(39.7)^2}{8} = 2.31875$

$S_{yy} = 275.47 - \dfrac{(49.5)^2}{9} = 3.22$

$s_p^2 = \dfrac{3.22 + 2.31875}{15} = 0.36925$

$$T = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{\sqrt{s_p^2\left(\frac{1}{n_x} + \frac{1}{n_y}\right)}}$$

$$= \frac{\frac{39.7}{8} - \frac{49.5}{9}}{\sqrt{0.36925\left(\frac{1}{8} + \frac{1}{9}\right)}} = -1.820$$

critical value $\rightarrow t_{0.95,15} = -1.753$

$-1.820 < -1.753 \rightarrow$ reject H_0

Question 5(c) is printed on the next page.

sufficient evidence that mean time taken by machine X is less than mean time for machine Y

(c) In general, would you expect the conclusions from the tests in parts **(a)** and **(b)** to be the same? Give a reason for your answer. [1]

t-test → assumes a normal distribution
→ equal variances
→ but this may not be true
→ therefore the two results may not be the same

Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

BLANK PAGE

BLANK PAGE

BLANK PAGE